9 Light

Light is a form of energy that help us to see things around us. The Sun is the main source of natural light on Earth.

Learn about

- Sources of light
- Luminous and non-luminous objects
- Properties of light
- How do we see through things?
- How are shadows formed?

Sources of Light

Objects that give out (emit) light are called sources of light. Sources of light can be of two types—natural and artificial.

Natural sources *of light are those natural objects that give off light on their own and are not made by man.* Sun is the main natural source of light on the Earth. Other examples are the stars and fire. The moon provides light at night but does not have its own light. It reflects the Sun's light that falls on it. Stars are so far away from us that they provide very little light to us. Some insects like glow-worms and fireflies also emit their own light.

Sun as a Natural Source of Light

Sunlight is necessary for life. During the daytime, we get light from the Sun.

Plants need sunlight for photosynthesis and flowering.

Think and Discuss

What will happen if there is no sunlight?

Fact File

Light coming from the Sun takes more than 8 minutes to reach the Earth.

Sun is a natural source of light.

Artificial sources *of light include those objects that emit light and are made by man for their use.* Some of the artificial sources of light use a cell or battery to function. Examples of artificial sources of light are electric bulbs and tubes, earthen lamps, kerosene lamps, torches, and wax candles.

Artificial sources of light

▶ Luminous and Non-luminous Objects

Objects that give out light of their own are known as ***luminous objects.*** Sun, an electric bulb, a candle,a lamp, and a firefly are some examples of luminous objects.

Objects that do not have their own light are called ***non-luminous objects.*** A tree, chair, cup, toy, book, and the moon are some examples of non-luminous objects.

Luminous objects

Non-luminous objects

Fact File

Thomas Alva Edison was an American inventor who invented many devices such as phonograph, motion picture camera, and electric light bulb.

REMEMBERING UNDERSTANDING

Questions

Give two examples of the following.

1. Natural sources of light ....
2. Artificial sources of light ....
3. Luminous objects ....
4. Non-luminous objects ....

Properties of Light

Light shows the following properties.

- Light is a form of energy.
- It always travels in a straight line. Let us understand this property with the help of the following activity.

Activity

Aim: To show that light travels in a straight line

Materials required: Three thick cards of equal size, modelling clay, a torch, a hole puncher, ruler, and a pencil

Procedure:

1. Number the cards as I, II, and III.
2. Punch a hole at the centre of each card. To identify the centre accurately, draw lines joining the opposite corners on each card. The point where the lines intersect will be the centre of the card. This way, the centre of the three cards will be at the same position.
3. Fix the three cards on a table (with the help of clay). Make sure there is a wall behind card III. The cards should be placed at an equal distance from each other. The holes of the cards should lie in a straight line.
4. Now, switch on the torch and place it in front of card I at a distance. Record your observations.
5. After this, gently move card II such that its hole is shifted. Record your observations.

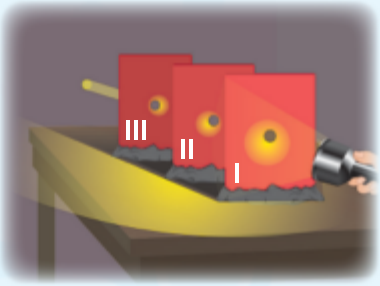


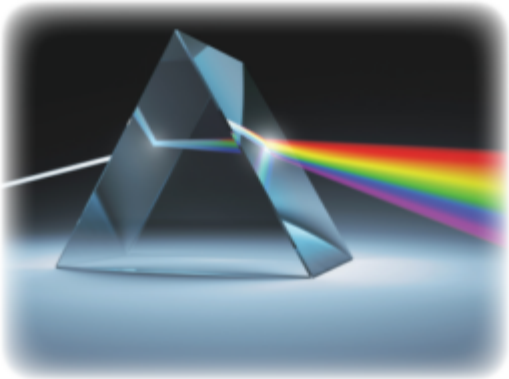
Fig. A

Observation: When the holes of all the cards are aligned, the light passes through all three holes and falls on the wall as a circular patch (Fig. A). When card II is moved, all three holes do not lie in a straight line. So, no circular patch appears on the wall this time.

Conclusion: Light travels in a straight line.

- Light travels in all directions from its source.
- *When light falls on the surface of an opaque object, it bounces back. This is known as* ***reflection of light****.* When the reflected rays of light reach our eyes, we are able to see the object.
- *When light passes from one medium to another, bending of light takes place at the boundary of the two media (medium refers to air, water, or some other material), and the object appears to be bent or broken. This is known as* ***refraction of light****.*

Straws appear bent when immersed in water.



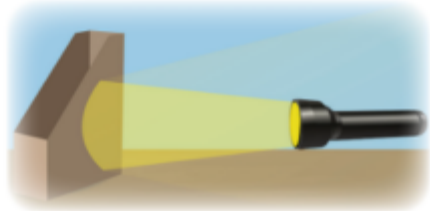
Splitting of light through a prism

Formation of rainbow due to splitting of light

- **Visible light** can be split into seven colours—violet, indigo, blue, green, yellow, orange, and red (**VIBGYOR**). This can be seen when light passes through a prism. The formation of a rainbow also takes place due to the splitting of light.

Light travels very fast, at a speed of 3×10^8 metre per second. Speed of light is considered to be the fastest in the universe.

How Do We See Through Things?



You can see through clean water or glass but not through your science book. This happens because different materials allow different amounts of light to pass through them.

Materials such as wood and cardboard do not allow any light to pass through them. Such materials are said to be ***opaque****.* The clothes we wear and the fruits we eat are opaque.

Wood (opaque)

Materials such as butter paper and coloured glass allow only some light to pass through them. Such materials are said to be ***translucent****.* Sunglasses are also made of translucent objects. Tracing paper, fine cloth, and ground glass are translucent objects. Translucent materials may be tinted (lightly coloured) or colourless.

Coloured glass (translucent)

Materials such as clear water, cellophane, and glass allow light to pass through them completely. Such materials are said to be ***transparent****.* Spectacles and windowpanes are made of transparent materials.

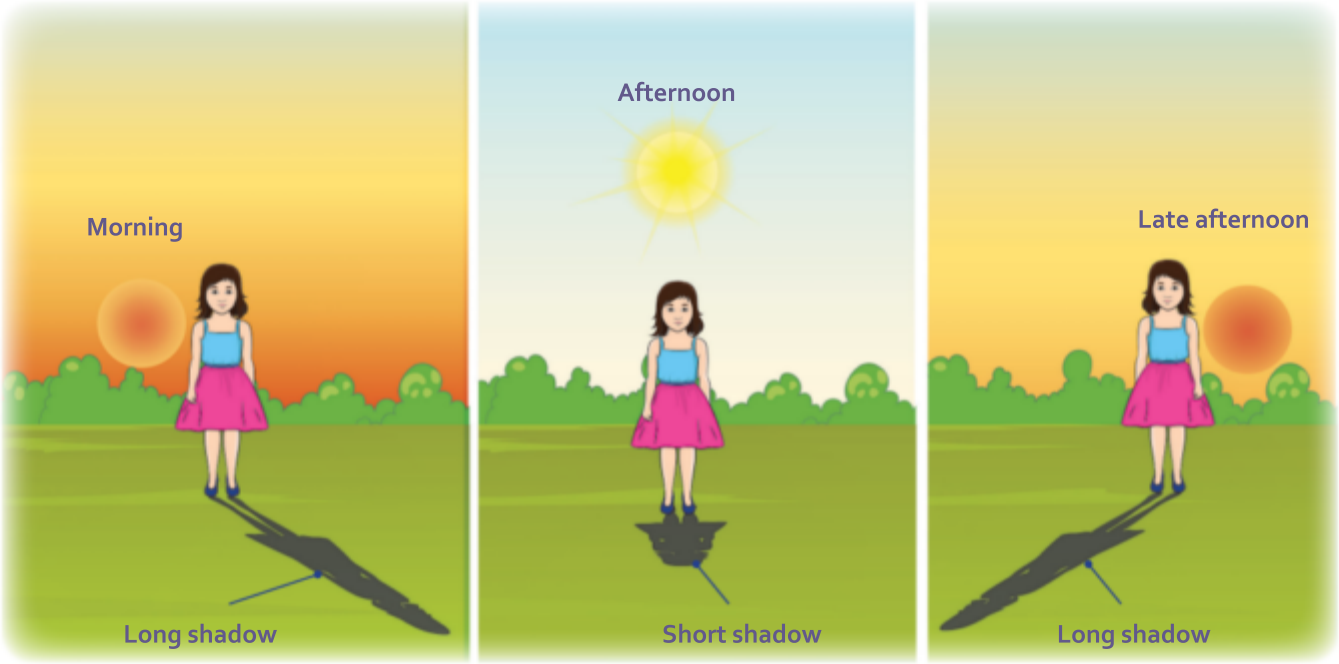
Glass (transparent)

Uses of transparent, translucent, and opaque objects We select transparent, translucent, or opaque objects based on whether we want to see through it fully (transparent), partially (translucent), or not at all (opaque). For example, the windows of our drawing rooms are mostly transparent glasses. Sunlight can enter through it, and we can see outside through it too. However, our bathroom window is frosted glass[1], which is translucent in nature. Thick curtains or blinds are opaque objects that block light, and thus they are opaque.

▶ How are Shadows Formed?

Light travels in a straight line. *When an opaque object comes in front of light, it blocks the light and forms a dark patch called a* ***shadow***. A shadow is always formed on the opposite side of the light. When you block sunlight, the shadow of your body is formed. When you move, your shadow moves along with you.

The sun appears to move across the sky from morning to evening. Accordingly, the shape and the length of the shadows change as well. Shadows are longest in the morning and in the late afternoon. They are shortest at noon when the sun shines overhead.



Sun and shadows

[1]**frosted glass:** a rough surface, so that it is difficult to see through

Questions

REMEMBERING UNDERSTANDING

Fill in the blanks with suitable words.

1. Light is a form of ..... .
2. When light falls on an ..... object, it bounces back. It is called ..... of light.
3. Rainbow formation take place due to ..... of light.
4. Frosted glass is a/an ..... object.
5. The shadow formed is the shortest at ..... when the sun is ..... .

Activity

TP SCIENTIFIC SKILLS EXPERIENTIAL LEARNING

Aim: To observe the change in the length of the shadow during the day

Materials required: Stick, watch, measuring tape, notebook, and pencil

Procedure:

1. Set up a stick in the middle of a playground on a bright sunny day.
2. Measure the length of the shadow of the stick in the morning, at noon, and again during late afternoon with the help of a measuring tape.

Observation: Record your observations in the table given below.

Time of the day	Length of the shadow

Case Study

Rakesh, a middle-aged man, has recently shifted to Kolkata for work. He is an early riser, so it is usually dark when he gets up. He has to switch on the LED light to get ready. He bought a house near a beach, so he prefers going there to watch the sun rise. One night, while returning from his office tour via the Howrah Express train, he saw many glow-worms shining brightly outside the window of his train.

a. Classify the sources of light used in the above passage as natural or artificial:

   i. LED bulb

   ii. Sun

   iii. glow-worms

b. Name two other natural and artificial sources of light.

Wrap Up

- Objects that give out (emit) light are called sources of light. These can be of two types—natural sources and artificial sources.
- Luminous objects are those that emit light of their own. For example, the sun, candles, and lamps.
- Non-luminous objects are those that do not have their own light. For example, chairs, cups, and the moon.
- When light falls on the surface of an opaque object, it bounces back. This is known as reflection of light.
- When light passes from one medium to another, bending of light takes place at the boundary of the two media. This is known as refraction of light.
- Visible light can be split into seven colours–violet, indigo, blue, green, yellow, orange, and red.
- Different materials allow different amounts of light to pass through them.
- A shadow is always formed on the opposite side of the source of light.

Exercises

SECTION I

APPLYING REMEMBERING UNDERSTANDING

A Choose the correct option.

1. Which of the following is a natural source of light?
   a. Glow-worm b. Wax candle
   c. Tube light d. Kerosene lamp
2. A candle is a ..... .
   a. non-luminous object b. luminous object
   c. natural source of light d. not a source of light
3. Visible light can be split into ..... colours.
   a. eight b. seven c. five d. six
4. Wood and cardboard are ..... objects.
   a. transparent b. opaque c. translucent d. luminous
5. A shadow is a ..... patch.
   a. light b. dark c. blue d. white

B Assertion and Reasoning questions.

1. **Assertion (A):** Table is a transparent object.
   Reason (R): Transparent objects do not allow light to pass through them.
   a. Both A and R are True b. Both A and R are False
   c. A is True and R is False d. A is False and R is True
2. **Assertion (A):** The length of a shadow is the shortest at noon.
   Reason (R): A shadow is always formed on the opposite side of the object.
   a. Both A and R are True b. Both A and R are False
   c. A is True and R is False d. A is False and R is True

C Choose the correct word to fill in the blank.

1. The ..... (Moon/Sun) is the main source of light on Earth.
2. A kerosene lamp is an example of a/an ..... (artificial/natural) source of light.
3. A glow-worm ..... (does not emit/emits) light.
4. Light always travels in a ..... (straight/zigzag) line.
5. An opaque object ..... (allows/does not allow) light to pass through it.

D **Write *T* for True and *F* for False. Correct the False statements.**

1. Table is a luminous object.
2. Light helps us to see the things around us.
3. Stars are non-luminous objects.
4. We can see a pencil bent when placed in a glass of water.
5. Light is a form of energy.
6. Shadows are longest at noon.

E **Give two examples of the following.**

1. Transparent objects ....
2. Translucent objects ....
3. Opaque objects ....

SECTION II

APPLYING ANALYSING

F **Short answer questions.**

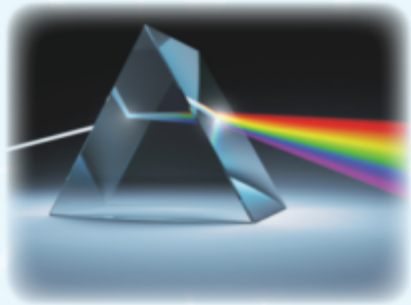
1. Define 'source of light'.
2. Why can't we see anything through a book?
3. List two uses of transparent objects.
4. What are the seven colours in which light splits when it passes through a prism?

G **Long answer questions.**

1. List the properties of light.
2. Differentiate between reflection and refraction of light.
3. Differentiate between a table and a glass based on their capability to allow light to pass through them.
4. Arpita was hanging the clothes for drying in the morning. While hanging, she noticed the long shadow of the clothes. When she came back to collect the clothes at noon, the length of the shadow reduced. Explain the reason for the change in the length of the shadow.

Picture Study

Observe the given picture carefully and answer the following questions.



1. Name the object shown.

   .....

2. Name the colours formed, when white light passes through it.

   .....

3. What is this phenomenon known as?

   .....

4. Name the practical application of this phenomenon around us.

   .....

My Learning Corner

A Think about

1. Name an object that is luminous as well as opaque. Suggest another object that is luminous and transparent.
2. If a wooden stick is immersed in water, it appears bent. Explain.

B Try out

1. You have studied in Social Studies that days and nights are caused due to the rotation of the Earth.

   As the Earth rotates, one-half of it that faces the Sun gets sunlight and so has day, while the other half that does not get sunlight has night (darkness). Try to explain it with an example of two countries—pair India and United States of America.

2. Perform the following activity to find out whether different objects in our surroundings are transparent, translucent, or opaque.

   Materials required: A glass, a wooden object, cellophane paper, a white sheet of paper, a piece of cardboard, oil, cotton, pencil, and a source of light

   Procedure:

   a. Take a glass and hold a pencil behind it, as shown alongside. Try to see through each material. Can you see the pencil?

b. Now, take a white sheet of paper and hold a pencil behind it. Try to see through it. Can you see through it?

c. Put 2 or 3 drops of oil on the white sheet of paper and smear it properly with the help of a cotton ball. Now try to see the pencil through this paper. Can you see the pencil now?

d. Repeat this activity with other materials.

Observation:

I was able to see through ..... .

I was not able to see through ..... .

I was able to partially see through .....

Conclusion:

..... are transparent objects.

..... are opaque objects.

..... are translucent objects.

3. Take a wooden scale and using a torch, cast the shadow of the wooden scale on a wall.

a. First, move the torch closer to the wooden scale, and observe the length of the shadow (Fig. A).

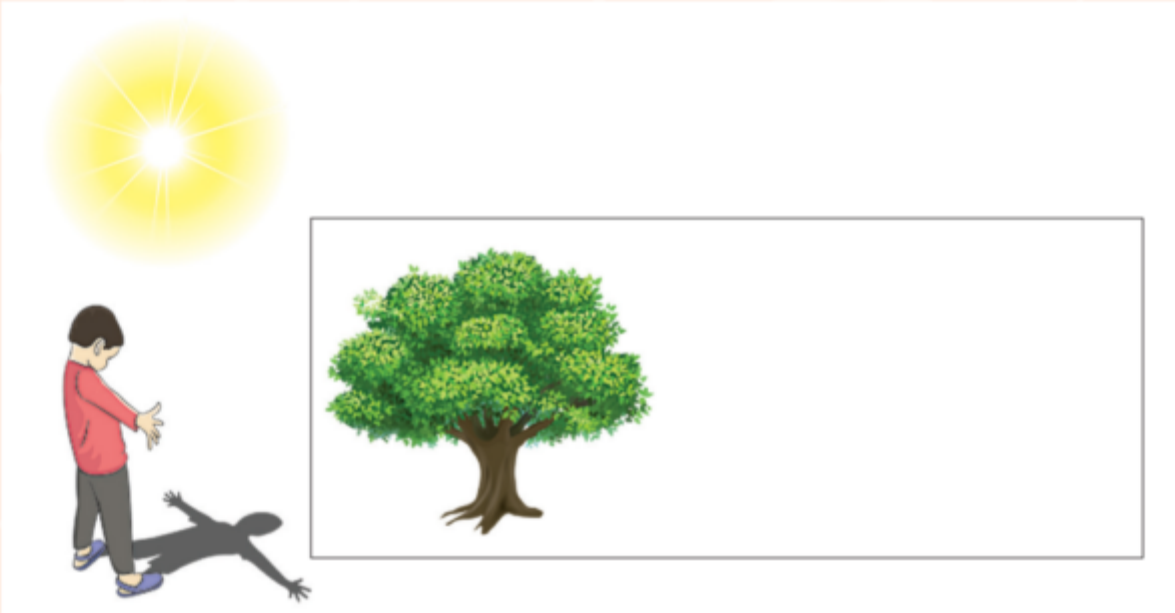
b. Now move the torch away from the wooden scale, and notice the change in it's length (Fig. B).

Fig. A **Fig. B**

Complete the given table based on the activity you have performed.

	Distance moved by the torch	Length of the shadow (increase/decrease)
1	Closer to the wooden scale	.....
2	Away from the wooden scale	.....

4. Look at the picture of a boy and his shadow. Now try to draw/complete the shadow of the tree given in another box. Also, colour it.



Self-Assessment

Now that you have completed the chapter, score each of the following tasks from 1 to 5 to indicate how well you can do them.

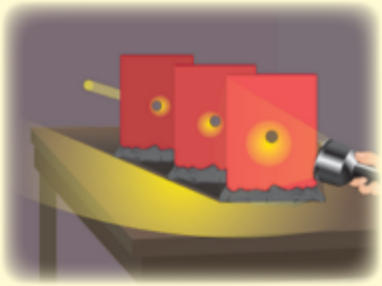
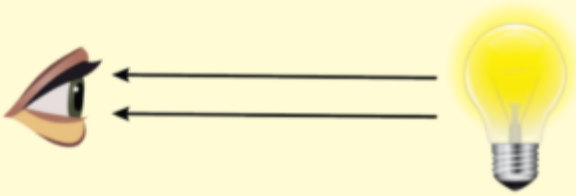
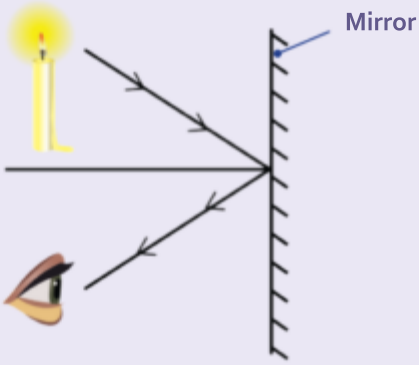
Score 5 = I can definitely do this.

Score 1 = I cannot do this yet.

I can...	My score
• explain the various sources of light.	
• differentiate between luminous and non-luminous objects.	
• list the properties of light.	
• explain how we see through things.	
• explain how shadows are formed.	
• perform different activities based on light.	

Worksheet

Which property of light do each of the following pictures depict?

	Pictures	Properties of Light
1.		....
2.		....
3.		....
4.		....